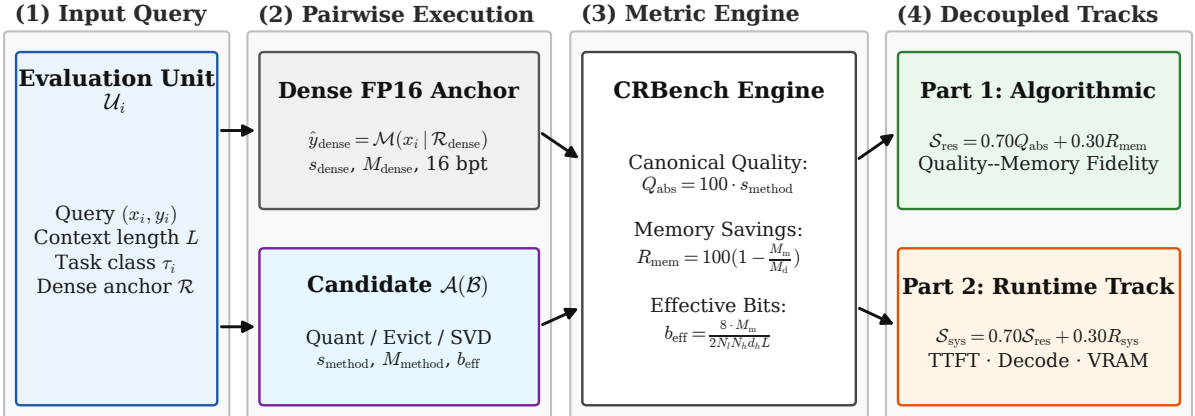
(1) Input Query
Evaluation Unit
$\mathcal{U}_i$
Query (x_i, y_i)
Context length L
Task class τ_i
Dense anchor $\mathcal{R}$
(2) Pairwise Execution
Dense FP16 Anchor
$\hat{y}_{\text{dense}} = \mathcal{M}(x_i \mid \mathcal{R}_{\text{dense}})$
s_{dense}, M_{dense}, 16 bpt
Candidate $\mathcal{A}(\mathcal{B})$
Quant / Evict / SVD
s_{method}, M_{method}, b_{eff}
(3) Metric Engine
CRBench Engine
Canonical Quality:
$Q_{\text{abs}} = 100 \cdot s_{\text{method}}$
Memory Savings:
$R_{\text{mem}} = 100(1 - \frac{M_{\text{m}}}{M_{\text{d}}})$
Effective Bits:
$b_{\text{eff}} = \frac{8 \cdot M_{\text{m}}}{2N_l N_h d_h L}$
(4) Decoupled Tracks
Part 1: Algorithmic
$\mathcal{S}_{\text{res}} = 0.70Q_{\text{abs}} + 0.30R_{\text{mem}}$
Quality--Memory Fidelity
Part 2: Runtime Track
$\mathcal{S}_{\text{sys}} = 0.70\mathcal{S}_{\text{res}} + 0.30R_{\text{sys}}$
TTFT · Decode · VRAM